

Tutorial 6

1. For the NMOS, given $N_D = 10^{20} \text{cm}^{-3}$, $N_A(\text{sub}) = 10^{16} \text{cm}^{-3}$, $W = 10 \mu\text{m}$, $L = 1.5 \mu\text{m}$, $L_{SD} = 5 \mu\text{m}$, $x_j = 0.4 \mu\text{m}$ determine
 - a. The drain diffusion capacitance for $V_{DB}=5\text{V}$ and $V_{DB}=2.5\text{V}$.
 - b. The overlap capacitance between gate and drain for $t_{ox}=200\text{\AA}$ and $L_{OV}=0.25\mu\text{m}$
2. A CMOS process has the following parameters - $V_{Tn} = 0.8 \text{V}$, $V_{Tp} = -1.0\text{V}$, $k'_n = 2k'_p = 120 \mu\text{A/V}^2$. The minimum L and W in the technology are $0.6 \mu\text{m}$ and $1.2 \mu\text{m}$ respectively. Assume that the combined load capacitance is 0.3pF and the input to the gate is an ideal step. Design a CMOS inverter to meet the following specifications
 - a. $V_{TH} = 1.5\text{V}$ for $V_{DD} = 3\text{V}$
 - b. Propagation delays $\tau_{pHL} \leq 0.2\text{ns}$ and $\tau_{pLH} \leq 0.15\text{ns}$
 - c. A falling delay of 0.35ns for an output transition from 2V to 0.5V
3. For a CMOS inverter fabricated in a $0.13 \mu\text{m}$ process with $V_{DD} = 1.2\text{V}$, $V_{Tn} = -V_{Tp} = 0.4\text{V}$, $k'_n = 4k'_p = 430 \mu\text{A/V}^2$. The inverter is designed with $(W/L)_n = 1.5$ and $(W/L)_p = 3$. Find τ_{pHL} , τ_{pLH} and τ_p , when the load capacitance is 10fF . Use the method of average current.